



Design and implementation of an open-source BDS-3 B1C/B2a SDR receiver

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Abstract

GNSS software-defined radio (SDR) receiver has been and will continue to be a tremendous research enabler given its flexibility and GNSS modernization as well as improvements to complimentary technologies. An open-source suite of GNSS SDRs capable of post-processing all open-service GNSS signals has been developed by the GNSS Lab at the University of Colorado, Boulder. As the latest expansion, processing capabilities for the B1C/B2a signals of the third-generation BeiDou navigation satellite system (BDS-3) are incorporated into this SDR package. To provide a basic implementation framework for GNSS community, separate or joint processing of the data and pilot channels are realized in the B1C/B2a SDR; and both narrowband and wideband tracking modes are implemented specifically for B1C pilot channel. Soon after the launch of the first two BDS-3 satellites, the B1C/B2a signals have been captured and the initial tracking results have been obtained. We describe the design strategy and implementation of the BDS-3 B1C/B2a SDR and report the processing results. The emphasis is placed on the B1C processing due to the novelty and complexity of the quadrature multiplexed binary offset carrier modulation employed by B1C.

Keywords BDS · Open source · RMS bandwidth · QMBOC · Interoperability

Introduction

The Chinese BeiDou Navigation Satellite System (BDS) has started its remarkable expansion from a former regional navigation capability to a global coverage through the construction of the third-generation (BDS-3) system. This is made possible after the test and verification of five in-orbit-validation satellites for the core system technologies of BDS-3 in 2015 and 2016. BDS-3 satellites feature higher-performance rubidium and hydrogen atomic clocks and improved signal-in-space (SIS) accuracy compared to BDS-2 satellites. They also feature inter-satellite links, better compatibility with other navigation systems, and offer search-and-rescue services. Among the four global navigation satellite systems (GNSS), BDS-3 is the only one that mixes geostationary earth orbits (GEO), inclined geosynchronous satellite orbits

(IGSO) and medium earth orbits (MEO) satellites, and integrates radio navigation satellite service (RNSS), radio determination satellite service (RDSS), and satellite-based augmentation system (SBAS) services into a single constellation (Kaplan and Hegarty 2017). The first pair of BDS-3 satellites was successfully launched on Nov 5, 2017. The full operational capability with more than 30 BDS-3 satellites is expected to be achieved by the end of this decade, according to the China Satellite Navigation Office (CSNO) (Shen 2017). In addition to the legacy B1I and B3I signals transmitted by BDS-2 satellites, the BDS-3 satellites transmit two new civil signals, namely B1C at 1575.42 MHz and B2a at 1176.45 MHz.

The B1C signal adopts the quadrature multiplexed binary offset carrier (QMBOC) modulation (Yao et al. 2010a) which shares the same carrier frequency and normalized power spectral density (PSD) with the composite BOC (CBOC) modulation of Galileo E1 OS signal and time multiplexed BOC (TMBOC) modulation of GPS L1C signal. The reception of L1C/E1 OS/B1C signals is expected to be a priority for multi-constellation GNSS civil receivers due to the interoperability between these three signals and their inherently superior performance. The B2a signal, which will

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replace the B2I signal transmitted by BDS-2 satellites, uses quadrature phase shift keying (QPSK) modulation (CSNO 2017b) and was intended to support interoperability with GPS L5 and Galileo E5a signals.

To fully exploit the SIS resources provided by the BDS-3 in the following years, it is certain that the receiver developers and researchers will integrate BDS-3 signal processing capability into multi-constellation GNSS receivers. This demands in-depth analysis and algorithm development for the new BDS-3 signals. High adaptable and flexible nature of the continually evolving SDR (Shivaramaiah and Akos 2017; Hussain et al. 2016) makes it an ideal environment for rapid prototyping and testing of new receiver architectures and algorithms for BDS-3 as well as other GNSSs. To this end, the GNSS Lab at the University of Colorado (CU) at Boulder has developed a suite of GNSS SDRs capable of post-processing all open-service GNSS signals. These SDRs adopt the framework of a MATLAB-based GPS C/A SDR code which was composed by Borre and Akos (2005) and now has been widely spread and employed in education and research. All of these SDRs are available as open-source software for GNSS community at www.colorado.edu/lab/gnss. The Chinese beta-version interface control documents (ICD) for BDS-3 B1C and B2a signals was released on September 5, 2017 (CSNO 2017a). In response to this ICD, a new SDR for these two signals has been developed and added to the existing SDR capabilities.

The purpose of BDS-3 SDR is to provide basic processing methodologies for the B1C and B2a signals according to their specific structures and characteristics, thereby establishing a platform for algorithm research and modifications. The salient features of this SDR are as follows.

1. This SDR processes B1C/B2a intermediate frequency (IF) data end-to-end, starting from the signal acquisition through to position, velocity and time (PVT) computation.

2. Separate or joint processing of the data and pilot channels are implemented for B1C/B2a signals. For joint signal acquisition, the noncoherent combining strategy (Borio et al. 2009) is performed over data and pilot channels. In the joint tracking mode, tracking error estimates from data and pilot channels are optimally combined at the discriminator level for the delay lock loop (DLL) and Costas phase lock loop (PLL), and both narrowband (NB) and wideband (WB) correlation approaches can be implemented for the B1C pilot channel according to user configurations.

Soon after the launch of the first pair of BDS-3 satellites, B1C and B2a signals were captured and preliminary processing results reported in Li et al. (2018). A significant refinement of the previous algorithms and analysis has been made in this work. The design strategies are improved, implementation process and results from various processing modes are presented and analyzed in more detail. Emphasis is placed on the B1C processing, as the B2a processing is relatively easier to implement.

B1C signal model and characteristics

This section presents the signal model and discusses the PSD and root mean square (RMS) bandwidth (BW) of the B1C signal.

Signal model

B1C signal which features a BOC(1,1) modulation in the data channel and a QMBOC(6,1,4/33) modulation in the pilot channel is a new variant of the spectrum constrained MBOC signal family. The pilot QMBOC modulation multiplexes a BOC(1,1) component and a BOC(6,1) component in phase quadrature with the same spreading codes. Neglecting the noise term, the digital IF signals of B1C data and pilot channels can be presented as:

$$s_{d,n} = \sqrt{\xi_d \cdot C} \cdot D(nT_s) \cdot c_d(nT_s - \tau) \cdot sc_{11}(nT_s - \tau) \cdot \cos(2\pi f_{IF} nT_s + \theta) \quad (1)$$

and

$$s_{p,n} = \underbrace{\sqrt{\xi_{p61}} \cdot C \cdot C_{p2nd}(nT_s) \cdot c_p(nT_s - \tau) \cdot sc_{61}(nT_s - \tau) \cdot \cos(2\pi f_{IF} nT_s + \theta)}_{s_{p61,n}} - \underbrace{\sqrt{\xi_{p11}} \cdot C \cdot C_{p2nd}(nT_s) \cdot c_p(nT_s - \tau) \cdot sc_{11}(nT_s - \tau) \cdot \sin(2\pi f_{IF} nT_s + \theta)}_{s_{p11,n}} \quad (2)$$

with

$$sc_{m1}(nT_s - \tau) = \text{sign}\{\sin[2\pi \cdot m \cdot f_{\text{ref}} \cdot (nT_s - \tau)]\} \quad (3)$$

where $D(\cdot)$ is the navigation (NAV) message symbol, $c_d(\cdot)$ and $c_p(\cdot)$ are the primary codes of data and pilot channels, respectively, $C_{p2nd}(\cdot)$ is the secondary code of the pilot channel, $sc_{m1}(\cdot)$ is the sinusoidal subcarrier where $m = 1$ for BOC(1,1) and $m = 6$ for BOC(6,1), T_s is the sampling period, f_{IF} is the IF with Doppler, τ is the spreading code delay, $\emptyset$ is the carrier phase, $f_{\text{ref}} = 1.023$ MHz is the reference frequency, $\text{sign}\{\cdot\}$ is the signum function, $s_{p11,n}$ and $s_{p61,n}$ represent the BOC(1,1) and BOC(6,1) components of the pilot channel, respectively, C is the total power of the B1C signal, and ξ_d , ξ_{p11} and ξ_{p61} are the fractions of total B1C power in the components $s_{d,n}$, $s_{p11,n}$ and $s_{p61,n}$, respectively, and their ratio is $\xi_d : \xi_{p11} : \xi_{p61} = 11 : 29 : 4$.

The whole B1C signal, r_n , to be processed by baseband functions of the SDR is expressed as:

$$r_n = s_{d,n} + s_{p,n} + \eta(nT_s) = s_{d,n} + s_{p11,n} + s_{p61,n} + \eta(nT_s) \quad (4)$$

where $\eta(\cdot)$ is the additive white Gaussian noise with PSD, N_0 , over the one-sided front end (FE) BW β_r . It can be seen, the useful B1C signal consists of $s_{d,n}$, $s_{p11,n}$ and $s_{p61,n}$.

B1C is the most technically sophisticated, flexible and capable of all BDS civil signals to date. It achieves comprehensive optimization with advanced signal design technologies while offering receiver designer significant flexibility in exploiting the advanced capabilities provided by the SIS. For more evaluations of the B1C signal, refer to Yao and Lu (2011, 2016).

PSD and RMS bandwidth

PSD is the primary frequency-domain characteristic of GNSS signals that determines the shape of the autocorrelation function (ACF) of the spreading waveform through the duality principle based on the Fourier transform. Over infinite BW, the normalized PSDs with unite area of B1C data and pilot channels can be expressed as:

$$G_d(f) = G_{11}(f) \quad (5)$$

and

$$G_p(f) = \frac{29}{33}G_{11}(f) + \frac{4}{33}G_{61}(f) \quad (6)$$

with

$$G_{m1}(f) = f_{\text{ref}}^2 \left[\frac{\sin\left(\frac{\pi f}{2 \cdot m \cdot f_{\text{ref}}}\right) \cdot \sin\left(\frac{\pi f}{f_{\text{ref}}}\right)}{\pi \cdot f \cdot \cos\left(\frac{\pi f}{2 \cdot m \cdot f_{\text{ref}}}\right)} \right]^2 \quad (7)$$

where in $G_{m1}(f)$, $m = 1$ is for the BOC(1,1) PSD, and $m = 6$ is for the BOC(6,1) PSD. The normalized PSD of the whole B1C can be expressed as:

$$G_{\text{B1C}}(f) = \frac{1}{4}G_d(f) + \frac{3}{4}G_p(f) = \frac{10}{11}G_{11}(f) + \frac{1}{11}G_{61}(f) \quad (8)$$

In GNSS signal design and evaluation, the RMS BW is an important parameter that comprehensively measures the effect of PSD shape on code ranging accuracy. For a signal with arbitrary PSD, $G(f)$, its RMS BW after FE filtering is defined as:

$$\bar{\beta}_{\text{rms}} = \sqrt{\frac{1}{\gamma} \int_{-\beta_r/2}^{\beta_r/2} f^2 \cdot G(f) df} \quad (9)$$

where γ is the fraction of power remaining after bandlimiting to the FE BW, β_r :

$$\gamma = \int_{-\beta_r/2}^{\beta_r/2} G(f) df \quad (10)$$

The Cramér–Rao lower bound of code tracking error variance in white noise using any DLL discriminator design is (Yao and Lu 2016; Betz 2016):

$$\sigma_{\text{CRLB}}^2 = \frac{B_L (1 - 0.5B_L T)}{\xi \cdot (2\pi \bar{\beta}_{\text{rms}})^2 \gamma \cdot C/N_0} \quad (11)$$

where B_L and T are the noise BW and update time of DLL, ξ is the fraction of total B1C power in the component being tracked. When the early-late spacing of discriminator is

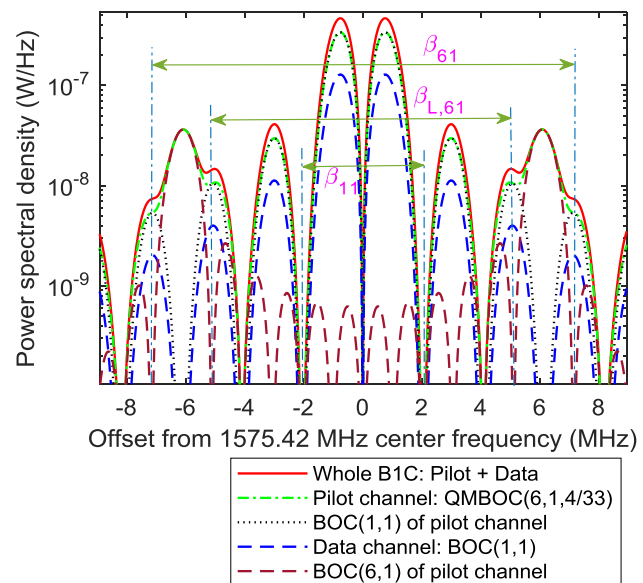


Fig. 1 PSDs for five combinations of the three B1C components

small and $C/N_0 > 30$ dB-Hz, the code tracking error variance can be approximated by σ_{CRLB}^2 (Betz 2016).

From (9) and (11), it can be inferred that, for a DLL with fixed B_L and T , the larger the RMS BW, $\bar{\beta}_{\text{rms}}$, the better is the potential ranging accuracy of the spreading waveform. In addition, larger $\bar{\beta}_{\text{rms}}$ generally implies a steeper main peak of the ACF and better multipath mitigation performance (Misra and Enge 2006; Avila-Rodriguez et al. 2008). Increasing both β_r and the high frequency portion within the spreading waveforms help to increase $\bar{\beta}_{\text{rms}}$.

The PSDs for five combinations of the three B1C components are shown in Fig. 1. All of the PSDs in Fig. 1 have two main lobes situated symmetrically about the center frequency. The BW of BOC-family signals is defined as the higher cutoff BW of the two main lobes (Xie 2009). In Fig. 1, except the PSD of $s_{p61,n}$ whose BW is $\beta_{61} = 14f_{\text{ref}} = 14.322$ MHz, the BWs of the other PSDs are $\beta_{11} = 4f_{\text{ref}} = 4.092$ MHz; and the lower cutoff BW of $s_{p61,n}$ is $\beta_{L,61} = 10f_{\text{ref}} = 10.23$ MHz. These three BW values identify three feature points in the RMS BW curves as depicted in Fig. 2.

Note that the RMS BW curve of $s_{p61,n}$ is not presented in Fig. 2, as it is generally impractical to use $s_{p61,n}$ only for positioning due to the small percentage of the power in the B1C signal. For comparison, the RMS BW curve of the BDS B1I signal is also plotted in Fig. 2. As can be seen, for fixed β_r , the $\bar{\beta}_{\text{rms}}$ of B1I is constantly smaller than the $\bar{\beta}_{\text{rms}}$ of all B1C components, although the code rate of B1I is twice as large as that of the B1C signal. It is interesting to

find that when $\beta_r \leq \beta_{L,61}$, the four RMS BW curves wholly coincide, which has a significant implication for B1C signal reception and processing: if the receiver FE BW is less than 10.23 MHz, no benefit can be obtained from processing $s_{p61,n}$ of the pilot channel in terms of code ranging accuracy. The RMS BWs of the full B1C and pilot-channel increase with much larger slopes at $\beta_r \in [\beta_{L,61}, \beta_{61}]$ corresponding to the frequency segment occupied by the two main lobes of $s_{p61,n}$, demonstrating a significant contribution of $s_{p61,n}$ to the RMS BWs. These observations provide some primary guidelines to BW selections for different approaches to B1C reception and correlation detailed in the next section.

Basic approaches to reception and correlation of B1C signal

In SDR receiver design, the complexity and cost of the sole hardware, i.e., the FE, are dictated by the signal reception strategy; and the computational load is mainly influenced by the correlator implementation. Various innovative reception and correlation technologies are expected to be developed to exploit the inherent capability of B1C fully. However, based on the BW, three basic approaches to reception and correlation of B1C signal can be identified, namely WB, NB, and sideband.

Due to the small proportion of the power of $s_{p61,n}$ in the B1C pilot channel, $s_{p61,n}$ can be seen as an enhancing component to $s_{p11,n}$. For high-end (e.g., survey) B1C receivers, the best possible performance in terms of code ranging accuracy and multipath resistance can be obtained by exploiting the WB B1C containing $s_{d,n}$, $s_{p11,n}$ and $s_{p61,n}$. Generally, at least the two main lobes of the PSD of $s_{p61,n}$ should be retained for WB reception of B1C. Therefore, β_r for B1C WB reception satisfies $\beta_{61} \leq \beta_r \leq \beta_{\text{Tx}}$, where $\beta_{\text{Tx}} = 32.72$ MHz is the B1C transmitting (Tx) BW as shown in Fig. 2.

For middle-end (e.g., navigation) B1C receivers that employ NB reception and correlation, $s_{p61,n}$ in the pilot channel is ignored, treating the QMBOC(6,1,44/33) as BOC(1,1). As shown in Fig. 2, the efficient selection of FE BW for NB reception of B1C is $\beta_{11} \leq \beta_r \leq \beta_{L,61}$. Compared to the WB approach, the NB reception of B1C can simplify the FE design and computational complexity of the SDR due to the reduced β_r and the associated sampling rate. Ignoring $s_{p61,n}$ can further reduce the correlator processing load in the pilot channel by more than half.

The sideband approach receives and processes either of the two main lobes of the BOC(1,1) PSD, which exploits the fact that each of these two main lobes contains all information necessary for receiver PVT computation. The advantage of this approach is that it treats the BOC(1,1) components like a pair of BPSK(1) signals at $1575.42 \text{ MHz} \pm f_{\text{ref}}$, which can significantly simplify FE structure and further reduce the

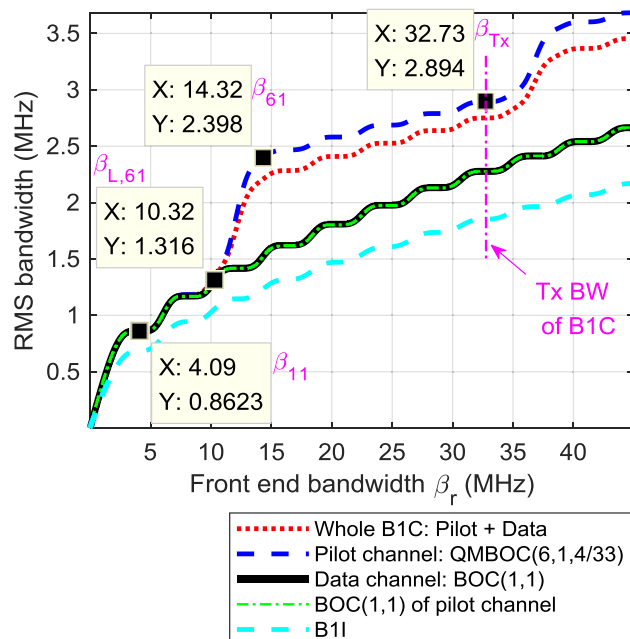


Fig. 2 RMS BWs for four combinations of the three B1C components and the B1I as a function of β_r

sampling rate, compared to the WB and NB approaches. In addition, for B1C receivers, sideband correlation in acquisition can take advantage of the width of the BPSK(1) ACF peak (Betz 2001; Fishman and Betz 2000). However, utilizing the sideband approach suffers at least 3 dB C/N_0 loss, as more than half of B1C power is filtered by the FE.

In this work, the B1C signal with WB PSD is received, and NB and WB correlation approaches are employed for baseband processing in the developed SDR.

Design and implementation of the SDR

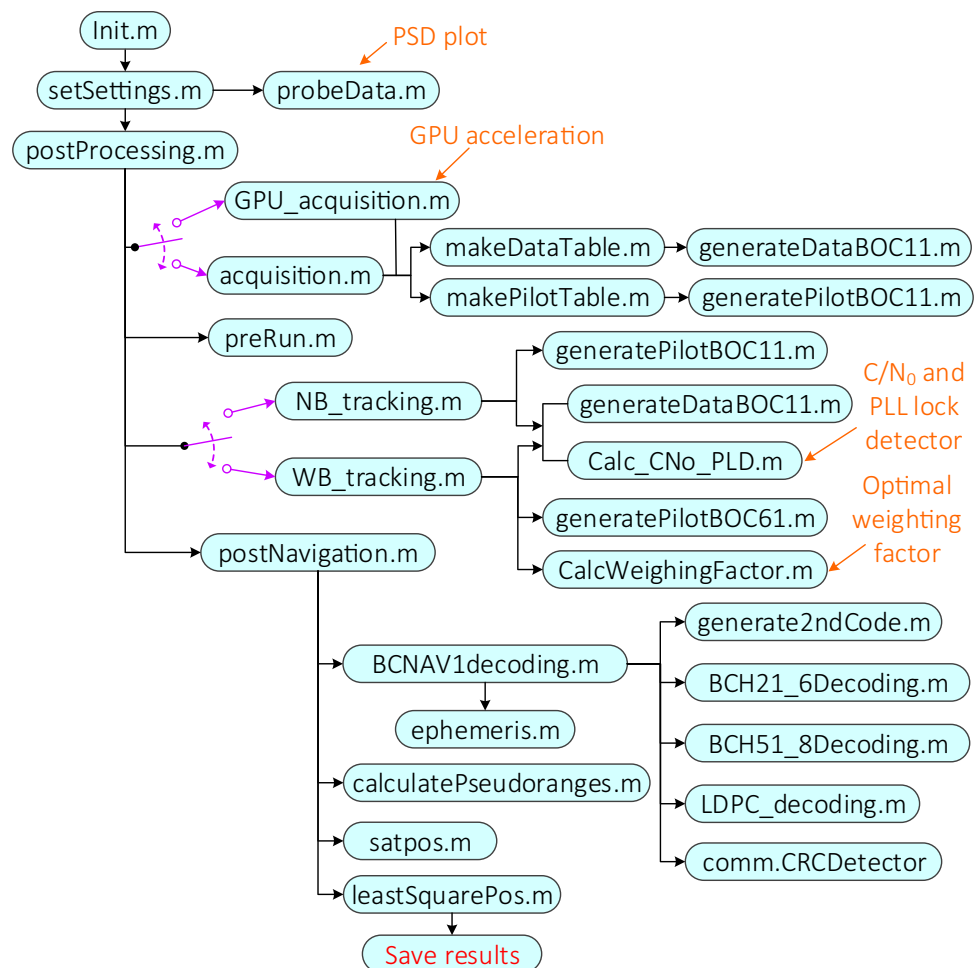
This section provides the basic methodologies for B1C baseband operations of the developed SDR. The ambiguity problem in signal acquisition and tracking of BOC modulated signals, which is not severe under low dynamics and normal C/N_0 conditions (Betz 2016), are not considered. Multiple methods in the literature (Lohan et al. 2017; Julien et al. 2007; Yao et al. 2010b) can be incorporated to solve this problem if needed by the SDR users.

High-level SDR structure and the captured signals

The BDS-3 B1C/B2a SDR adopts the framework composed by Borre and Akos (2005) for a GPS C/A SDR, with new algorithms implemented to adapt to the specific spreading modulations and NAV message structures of B1C and B2a signals. The high-level structure of the SDR for B1C processing is depicted in Fig. 3.

Soon after the launch of the first pair of BDS-3 satellites, we captured the B1C and B2a signals using a four-channel IF sampler (NT1065) from the Amungo Navigation. Figure 4 illustrates the signal PSD and time-domain plot of the captured B1C IF samples. In the PSD plot, only a small lobe centered at the IF can be observed due to the combined signal powers at 1575.42 MHz from all GPS satellites in view. In the time domain, no discernable structure of the B1C signals can be seen. However, the useful information buried within the noise of the received bandwidth can be extracted by baseband processing which is detailed in the following sections.

Fig. 3 High-level structure of the developed SDR for B1C signal processing (Li et al. 2018)



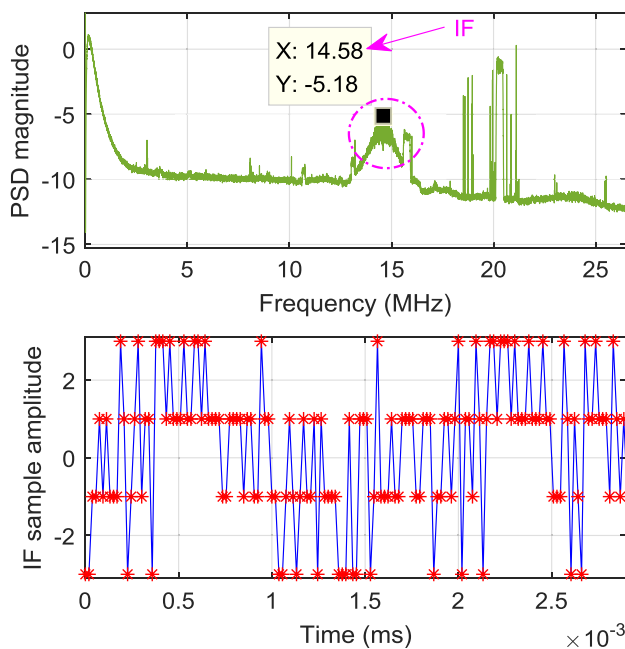


Fig. 4 PSD (top) and time-domain plot (bottom) of the collected IF samples

Signal acquisition

In signal acquisition stage only coarse estimations of the code delay and carrier frequency shift need to be provided, thus processing $s_{p61,n}$ of the pilot channel is typically not worth the complexity. Hence, a NB correlation approach is

employed in the coarse and fine acquisitions of the developed SDR. The core operation of coarse acquisition is to generate a matrix of the cross-ambiguity function (CAF) values by correlating the incoming signal with the local signal replicas whose code delay and frequency shift are searched over a maximum uncertainty region. It is well known that the sign transition of unknown message symbol and unsynchronized secondary code during the predetection integration time (PIT) can significantly degrade the processing gain of signal acquisition (Fortin et al. 2015; Foucras et al. 2014). This problem makes the B1C the most challenging signal to acquire in terms of sign transition probability, which is 50% for B1C as both its secondary code and message symbol duration match their primary code duration. To solve this problem, recognizing that the search over code delay dominates the operation count and storage of the coarse acquisition (Betz 2016), the circular correlation algorithm (Tsui 2005) based on zero padding is adopted in the SDR. The block diagram of the joint processing of data and pilot channels for B1C coarse acquisition is shown in Fig. 5.

Denote the B1C primary code length as $L_C = 10$ ms. As shown in Fig. 5, the length of the incoming IF signal segment used to do correlation is $2L_C$, which can ensure at least one complete primary code is contained. For either of data or pilot channels, the local samples of one primary code with BOC(1,1) subcarrier is padded with the same length of zeros, and all elements of the CAF matrix along each frequency bin can be calculated in parallel through circular correlation employing a set of discrete Fourier transform (DFT) and inverse DFT (IDFT) operations. Depending on

Fig. 5 Block diagram of the joint processing of data and pilot channels for B1C coarse acquisition

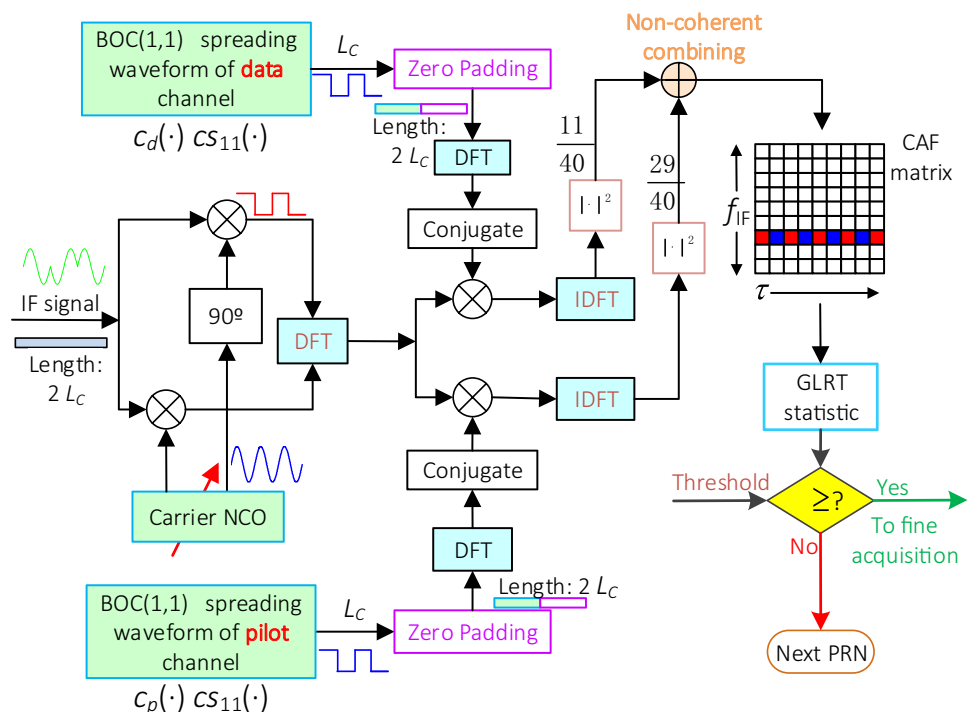
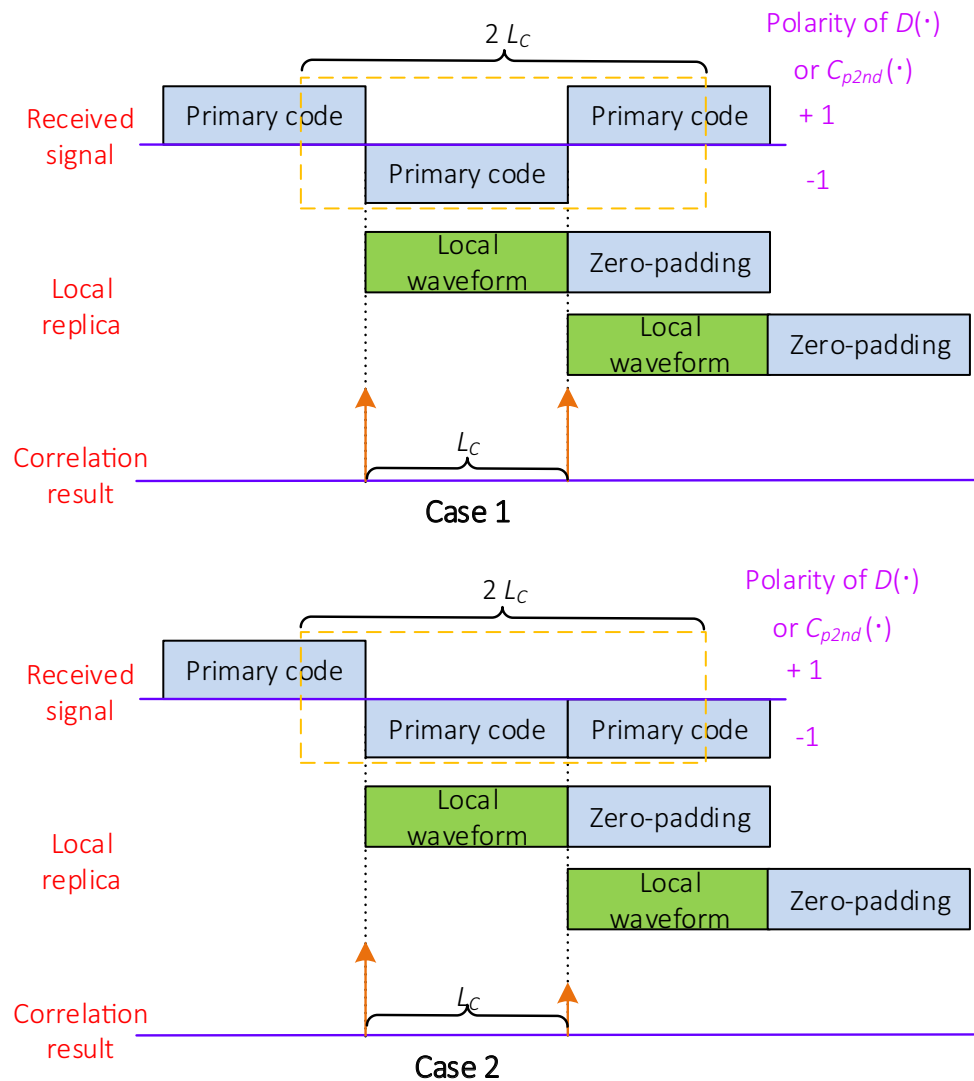


Fig. 6 Two possible correlation results of zero-padding circular correlation at the correct frequency bin



the relative polarities of $D(\cdot)$ and $C_{p2nd}(\cdot)$ contained in the incoming IF signal used to do correlation, there are two possible correlation results at the correct frequency bin for the data or pilot channel, as shown in Fig. 6. As can be seen, the effective PIT of the useful signal is L_C , while the actual DFT/IDFT length of circular correlation is doubled due to zero padding. To accelerate the coarse acquisition code, computationally efficient correlation algorithms, e.g., the double block zero padding or DBZP (Lin et al. 1999; Foucras et al. 2012) and the parallel time–frequency search algorithm (Stein 1981), can be easily incorporated into the existing code.

After the circular correlation operations for all possible frequency bins, the 2D CAF matrices from both channels are noncoherently combined with weighting factors according to the ratio between ξ_d and ξ_{p11} . The coarse acquisition result can be obtained by evaluating the sum CAF matrix. Several test statistic formation methods exist (Van Trees 2004;

Helstrom 2013; Geiger et al. 2012) for deciding whether the target satellite is in view or not. For simplicity of threshold setting, the generalized likelihood ratio test (GLRT) statistic (Ziemer and Peterson 2008) is employed, which has constant false alarm rate independent of the noise power. Once the GLRT statistic exceeds the predefined threshold, the coarse acquisition is concluded, and the fine frequency shift is then searched serially around the acquired coarse frequency bin with narrower frequency bin spacing.

The GLRT statistics of coarse acquisition from: (a) the separate processing of the data channel and (b) noncoherent combining of correlation results from both channels are compared in Fig. 7. Signals transmitted by the first two BDS-3 satellites with PRNs # 19 and # 20 are all acquired, and the GLRT statistic obtained from joint acquisition (bottom panel) is significantly larger than when only the data channel is used (top panel).

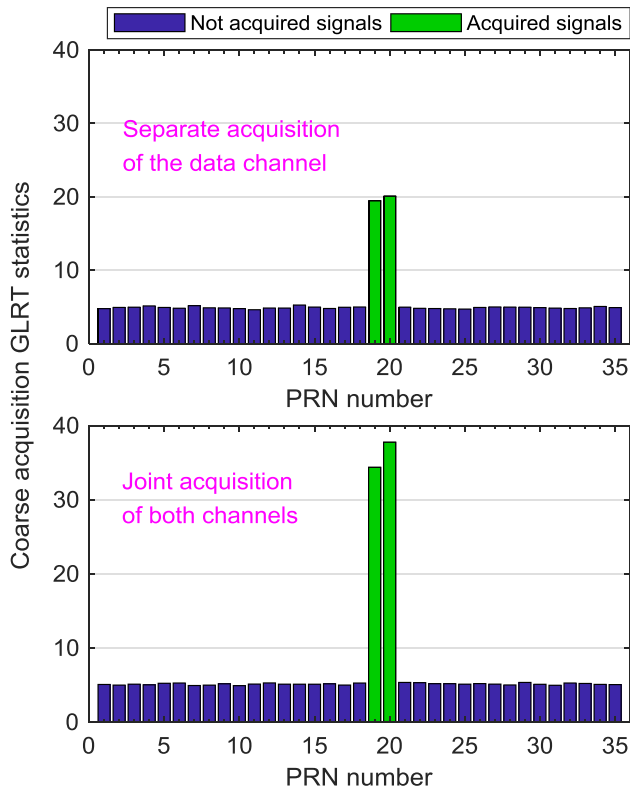


Fig. 7 GLRT statistics for the coarse acquisition of B1C signals

The correlation results for PRN #20 are shown in Fig. 8. In the top panel where the CAF envelope is portrayed, two peak groups of the CAF magnitudes signify presence of the signal; and the coordinates of the largest magnitudes within each peak groups reveal the rough code delay and frequency shift of the target signal, because the distance between the two largest magnitudes along the code delay dimension equals approximately the primary code length. The bottom panel shows the zoomed-in profile of the peak group having the maximum magnitude along the code delay dimension. The main peak, as well as two side peaks of the ACF of BOC (1,1) waveform, can be identified. However, in the circular correlation, the code delay resolution, equal to the IF sampling period, is adequate to distinguish the main peak from the two side peaks under normal C/N_0 conditions.

NB tracking mode

Using NB correlation approach, the operation of separate tracking of data or pilot channel is similar to that in the original L1 C/A SDR, if the ambiguous tracking problem is not considered. Therefore, only the joint tracking of the B1C signal is described herein, and its processing flow based on the NB correlation approach is portrayed in Fig. 9.

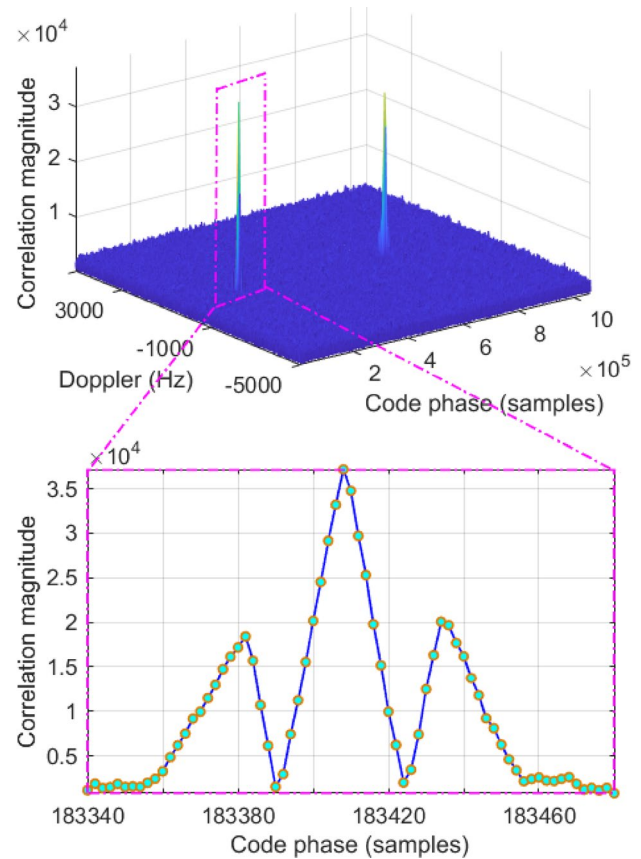


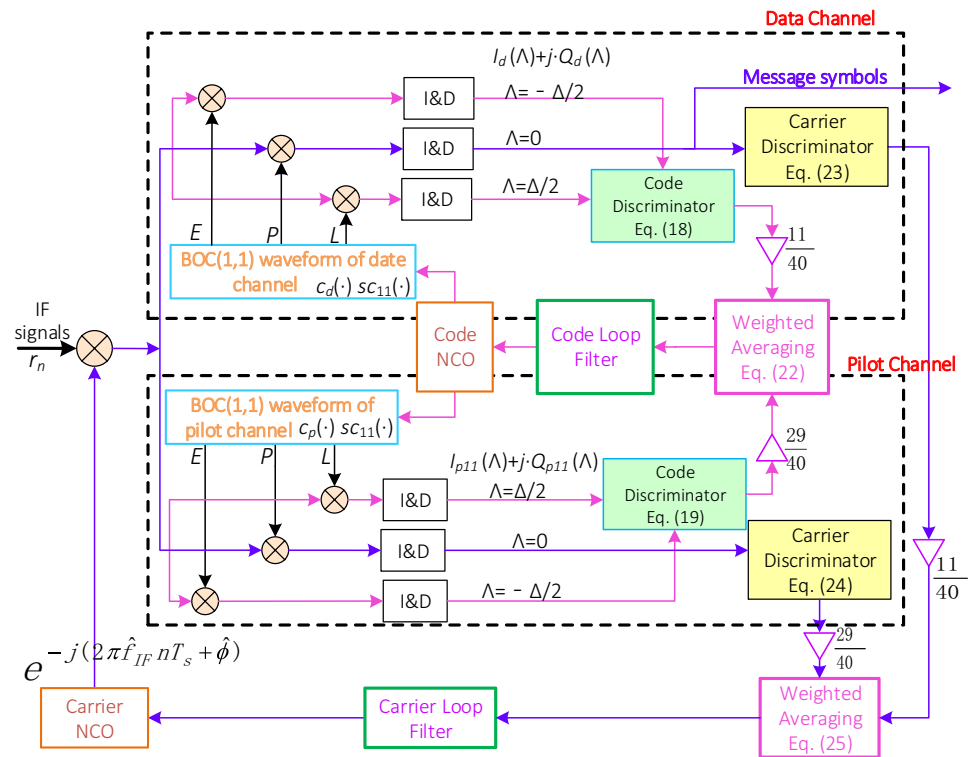
Fig. 8 Correlation results of coarse acquisition for PRN # 20 by non-coherent combining correlation results of both channels

For the pilot channel, denote the BOC(1,1) spreading waveform replica as $\hat{w}_{p11}(\Lambda) = c_p(nT_s - \hat{\tau} + \Lambda) \cdot sc_{11}(nT_s - \hat{\tau} + \Lambda)$, where $\hat{\tau}$ is the locally estimated code phase and Λ is the code-phase shift with respect to $\hat{\tau}$. Then, the complex correlation values for $s_{p11,n}$ are computed through the integration and dump operation:

$$I_{p11}(\Lambda) + j \cdot Q_{p11}(\Lambda) = \sum_{n=1}^M r_n \cdot e^{-j(2\pi\hat{f}_{IF}nT_s + \hat{\theta})} \cdot \hat{w}_{p11}(\Lambda) \quad (12)$$

where $M = T/T_s$ is the sample number within the PIT, T , $e^{-j(2\pi\hat{f}_{IF}nT_s + \hat{\theta})}$ is the complex carrier replica common to both channels, $\hat{f}_{IF}$ and $\hat{\theta}$ are the locally estimated frequency and data-channel phase of the carrier, respectively. For simplicity, assuming $\hat{f}_{IF} = f_{IF}$ and ignoring the noise term, $\eta(\cdot)$, due to orthogonality between $c_p(\cdot) \cdot sc_{11}(\cdot)$ and $c_d(\cdot) \cdot sc_{11}(\cdot)$, the results in (12) expressed in scalar form can be approximated as:

Fig. 9 Joint tracking of B1C data and pilot channels based on NB correlation approach



$$\begin{cases} I_{p11}(\Lambda) = K \left[\sqrt{\lambda} R_X(\delta\tau + \Lambda) \cos(\delta\theta) - \sqrt{1 - \lambda} R_{11}(\delta\tau + \Lambda) \sin(\delta\theta) \right] \\ Q_{p11}(\Lambda) = K \left[\sqrt{\lambda} R_X(\delta\tau + \Lambda) \sin(\delta\theta) + \sqrt{1 - \lambda} R_{11}(\delta\tau + \Lambda) \cos(\delta\theta) \right] \end{cases} \quad (13)$$

where $\delta\theta = \theta - \hat{\theta}$ is the data-channel carrier-phase tracking error, $\delta\tau = \tau - \hat{\tau}$ is the code-phase tracking error, $R_{11}(\cdot)$ is the ACF of $c_p(\cdot) \cdot sc_{11}(\cdot)$, R_X is the cross-correlation function between $c_p(\cdot) \cdot sc_{11}(\cdot)$ and $c_p(\cdot) \cdot sc_{61}(\cdot)$, and λ and K are:

$$\lambda = \frac{\xi_{p61}}{\xi_{p61} + \xi_{p11}} = \frac{4}{33} \quad (14)$$

and

$$K = \frac{1}{2} M \cdot C_{p2nd} \cdot \sqrt{(\xi_{p61} + \xi_{p11})C} \quad (15)$$

In NB processing, R_X is ignored and (13) becomes:

$$\begin{cases} I_{p11}(\Lambda) \approx -K \sqrt{1 - \lambda} R_{11}(\delta\tau + \Lambda) \sin(\delta\theta) \\ Q_{p11}(\Lambda) \approx K \sqrt{1 - \lambda} R_{11}(\delta\tau + \Lambda) \cos(\delta\theta) \end{cases} \quad (16)$$

Similar to the derivation of (13), the correlation values for $s_{d,n}$ can be obtained as:

$$\begin{cases} I_d(\Lambda) = \frac{1}{2} M \cdot D \cdot \sqrt{\xi_d} C R_d(\delta\tau + \Lambda) \cos(\delta\theta) \\ Q_d(\Lambda) = \frac{1}{2} M \cdot D \cdot \sqrt{\xi_d} C R_d(\delta\tau + \Lambda) \sin(\delta\theta) \end{cases} \quad (17)$$

where $R_d(\cdot)$ is the ACF of $c_d(\cdot) \cdot sc_{11}(\cdot)$.

The correlation values in (13), (16) and (17) are expressed in general form with parameter Λ . Assuming Δ is the early-late spacing of the DLL, the early, prompt and late correlation values can be produced by setting Λ to be $-\Delta/2$, 0 and $\Delta/2$, respectively. For joint tracking, the key is to combine these correlation values from both channels to form the composite tracking error estimate which continuously steers the numerically controlled oscillator (NCO) shared by both channels. In the developed SDR, this is done by the weight average of discriminator outputs.

For the DLL, the respective code tracking errors for $s_{d,n}$ and $s_{p11,n}$ are estimated through the popular normalized early-late magnitude (NELM) discriminators by:

$$\delta\hat{\tau}_d = \left(1 - \frac{\Delta}{2}\right) \frac{\sqrt{I_d^2(-\Delta/2) + Q_d^2(-\Delta/2)} - \sqrt{I_d^2(\Delta/2) + Q_d^2(\Delta/2)}}{\sqrt{I_d^2(-\Delta/2) + Q_d^2(-\Delta/2)} + \sqrt{I_d^2(\Delta/2) + Q_d^2(\Delta/2)}} \quad (18)$$

and

$$\delta\hat{\tau}_{p11} = \left(1 - \frac{\Delta}{2}\right) \frac{\sqrt{I_{p11}^2(-\Delta/2) + Q_{p11}^2(-\Delta/2)} - \sqrt{I_{p11}^2(\Delta/2) + Q_{p11}^2(\Delta/2)}}{\sqrt{I_{p11}^2(-\Delta/2) + Q_{p11}^2(-\Delta/2)} + \sqrt{I_{p11}^2(\Delta/2) + Q_{p11}^2(\Delta/2)}} \quad (19)$$

Assuming the weighting factor for $\delta\hat{\tau}_d$ is α_{nb} , the composite code tracking error can be expressed as:

$$\delta\hat{\tau}_{nb} = \alpha_{nb} \cdot \delta\hat{\tau}_d + (1 - \alpha_{nb}) \cdot \delta\hat{\tau}_{p11} \quad (20)$$

According to (11), the value of α_{nb} that minimizes the estimation error variance of $\delta\hat{\tau}_{nb}$ in white noise can be obtained as:

$$\alpha_{nb} = \frac{\xi_d \cdot \gamma_d \cdot (\bar{\beta}_{d,rms})^2}{\xi_d \cdot \gamma_d \cdot (\bar{\beta}_{d,rms})^2 + \xi_{p11} \cdot \gamma_{p11} \cdot (\bar{\beta}_{p11,rms})^2} \quad (21)$$

where the subscript d and p11 represent embodiments of the corresponding variables for $s_{d,n}$ and $s_{p11,n}$, respectively. Since $s_{d,n}$ and $s_{p11,n}$ have the same PSD, $\gamma_d = \gamma_{p11}$ and $\bar{\beta}_{d,rms} = \bar{\beta}_{p11,rms}$ holds. Therefore, according to the ratio between ξ_d and ξ_{p11} , equation (20) becomes:

$$\delta\hat{\tau}_{nb} = \frac{11}{40} \cdot \delta\hat{\tau}_d + \frac{29}{40} \cdot \delta\hat{\tau}_{p11} \quad (22)$$

The PLL operation is not affected by the BOC modulation once the spreading waveform of the IF signal is accurately wiped off by the local replica. Due to the unknown sign transition of $D(\cdot)$, the arctan discriminator, which is the most precise of Costas discriminators (Borre et al. 2007), is used to estimate the carrier-phase tracking error for $s_{d,n}$:

$$\delta\hat{\theta}_d = \tan^{-1} \left[\frac{Q_d(0)}{I_d(0)} \right] \quad (23)$$

As synchronization (sync) to the secondary code, $C_{p2nd}(\cdot)$, is performed after signal tracking in the developed SDR, the carrier-phase tracking error for $s_{p11,n}$ is also estimated by the arctan discriminator. According to (16), the estimation of carrier-phase tracking error for $s_{p11,n}$ is expressed as:

$$\delta\hat{\theta}_{p11} = \tan^{-1} \left[-\frac{I_{p11}(0)}{Q_{p11}(0)} \right] \quad (24)$$

Similar to (22), the weight average of $\delta\hat{\theta}_d$ and $\delta\hat{\theta}_{p11}$ can be obtained as:

$$\delta\hat{\theta}_{nb} = \frac{11}{40} \cdot \delta\hat{\theta}_d + \frac{29}{40} \cdot \delta\hat{\theta}_{p11} \quad (25)$$

Through (25), the estimation error variance of $\delta\hat{\theta}_{nb}$ in white noise can be minimized.

Figure 10 shows the joint tracking results based on NB correlation approach for the B1C signal of PRN # 20. In the top panel, the scatter pattern of I & Q prompt correlation

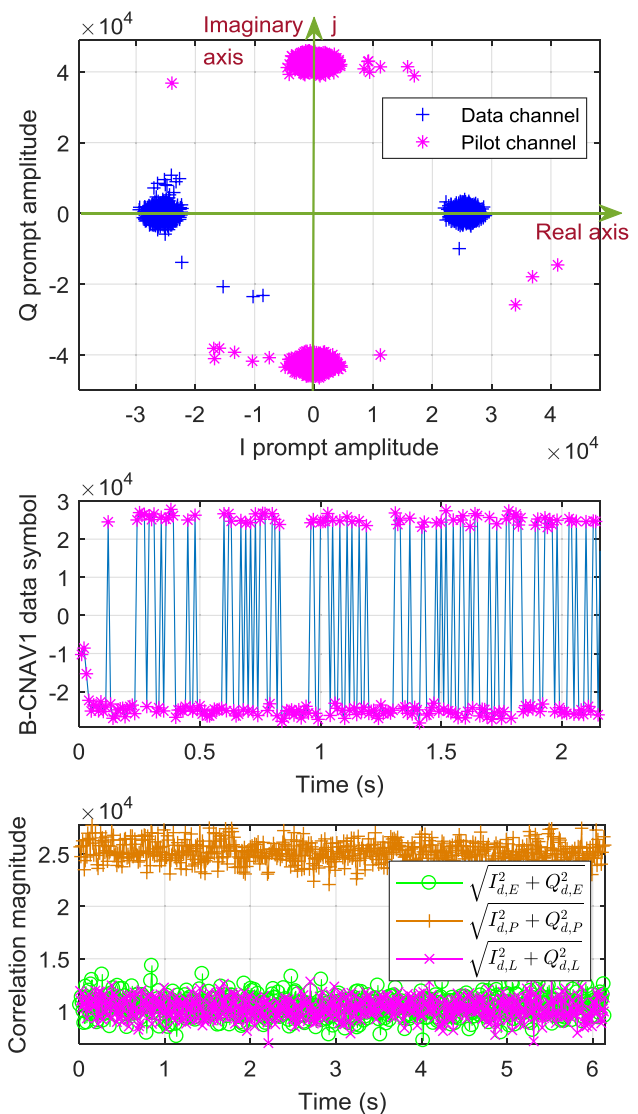
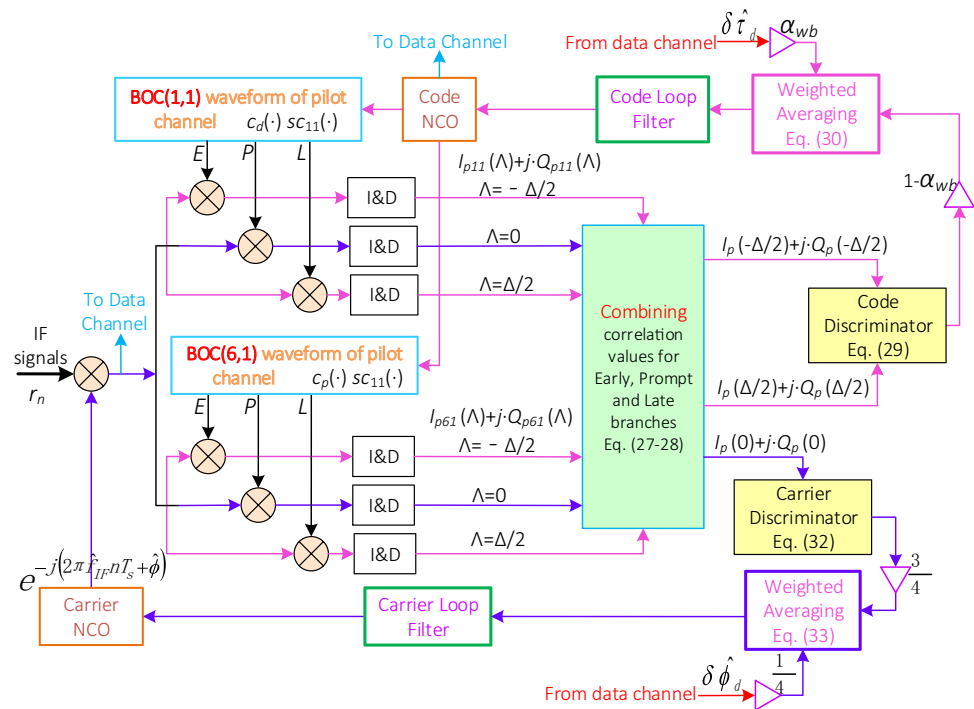


Fig. 10 Joint tracking results of the B1C signal based on NB correlation approach for PRN #20

Fig. 11 Pilot channel processing flow of the joint tracking based on WB correlation approach



values for both channels reflect the carrier-phase relationship between $s_{d,n}$ and $s_{p11,n}$ with modulation by $D(\cdot)$ and $C_{p2nd}(\cdot)$. In the middle panel, the negative and positive polarities of $D(\cdot)$ can be explicitly identified except at the beginning epochs when the DLL and PLL are in a transient state. In the bottom panel, the prompt correlation values are clearly separated from the early and late correlation values in magnitude. These results demonstrate stable and accurate tracking of the DLL and PLL of the SDR in NB tracking mode.

WB tracking mode

In joint tracking mode based on WB correlation approach, all the three components, $s_{d,n}$, $s_{d,n}$ and $s_{d,n}$, of B1C are processed; and the correlator operations for $s_{d,n}$ and $s_{d,n}$ are the same as in the NB tracking mode, while three sets of correlators for $s_{d,n}$ are added in the pilot channel. The processing flow of pilot channel in the joint tracking based on WB correlation approach is portrayed in Fig. 11.

Similar to (13), the correlation values for $s_{p61,n}$ can be obtained as:

$$\begin{cases} I_{p61}(\Lambda) = K \left[\sqrt{\lambda} R_{61}(\delta\tau + \Lambda) \cos(\delta\theta) - \sqrt{1-\lambda} R_X(\delta\tau + \Lambda) \sin(\delta\theta) \right] \\ Q_{p61}(\Lambda) = K \left[\sqrt{\lambda} R_{61}(\delta\tau + \Lambda) \sin(\delta\theta) + \sqrt{1-\lambda} R_X(\delta\tau + \Lambda) \cos(\delta\theta) \right] \end{cases} \quad (26)$$

where $R_{61}(\cdot)$ is the ACF of $c_p(\cdot) \cdot sc_{61}(\cdot)$. The composite correlation values for the complete pilot channel, $s_{p,n}$, can be found by combining (13) and (26) as:

$$\begin{aligned} I_p(\Lambda) &= \sqrt{\lambda} I_{p61}(\Lambda) + \sqrt{1-\lambda} Q_{p11}(\Lambda) \\ &= K \left[\lambda R_{61}(\delta\tau + \Lambda) + (1-\lambda) R_{11}(\delta\tau + \Lambda) \right] \cos(\delta\theta) \\ &= K \cdot R_{QMBOC}(\delta\tau + \Lambda) \cdot \cos(\delta\theta) \end{aligned} \quad (27)$$

and

$$\begin{aligned} Q_p(\Lambda) &= \sqrt{\lambda} Q_{p61}(\Lambda) - \sqrt{1-\lambda} I_{p11}(\Lambda) \\ &= K \left[\lambda R_{61}(\delta\tau + \Lambda) + (1-\lambda) R_{11}(\delta\tau + \Lambda) \right] \sin(\delta\theta) \\ &= K \cdot R_{QMBOC}(\delta\tau + \Lambda) \cdot \sin(\delta\theta) \end{aligned} \quad (28)$$

where $R_{QMBOC}(\cdot)$ is the CAF of $\sqrt{\lambda} c_p(\cdot) \cdot sc_{61}(\cdot) + j\sqrt{1-\lambda} c_p(\cdot) \cdot sc_{11}(\cdot)$. It can be seen that $I_p(\Lambda) + jQ_p(\Lambda)$ has the same characteristics as the conventional correlation values (Yao et al. 2010a).

In WB tracking mode, the NELM code discriminator output for the pilot channel is:

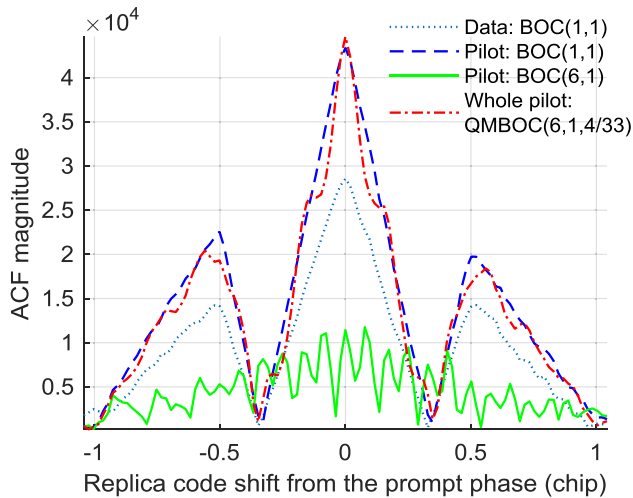


Fig. 12 ACF magnitude curves obtained from the WB tracking mode for PRN #20

$$\delta \hat{\tau}_p = \left(1 - \frac{\Delta}{2}\right) \frac{\sqrt{I_p^2(-\Delta/2) + Q_p^2(-\Delta/2)} - \sqrt{I_p^2(\Delta/2) + Q_p^2(\Delta/2)}}{\sqrt{I_p^2(-\Delta/2) + Q_p^2(-\Delta/2)} + \sqrt{I_p^2(\Delta/2) + Q_p^2(\Delta/2)}} \quad (29)$$

Note that to exploit the superior ranging capability of $s_{p61,n}$ fully, Δ should be kept below approximately 0.153 chip of the primary code (Betz 2016). Assuming the weighting factor for $\delta \hat{\tau}_p$ is α_{wb} , the composite code tracking error can be expressed as:

$$\delta \hat{\tau}_{wb} = \alpha_{wb} \cdot \delta \hat{\tau}_d + (1 - \alpha_{wb}) \cdot \delta \hat{\tau}_p \quad (30)$$

Similar to (21), α_{wb} can be obtained as:

$$\alpha_{wb} = \frac{11 \cdot \gamma_d \cdot (\bar{\beta}_{d,rms})^2}{11 \cdot \gamma_d \cdot (\bar{\beta}_{d,rms})^2 + 33 \cdot \gamma_p \cdot (\bar{\beta}_{p,rms})^2} \quad (31)$$

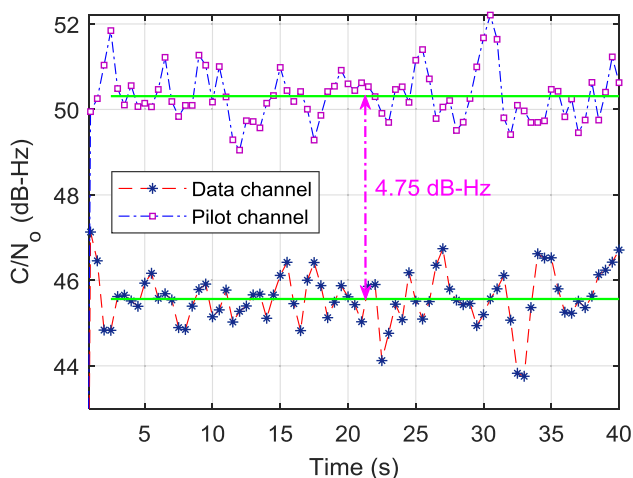


Fig. 13 C/N_0 estimations for PRN #20 in WB tracking mode

where γ_p is the power fraction of $s_{p,n}$ remaining after bandlimiting to β_r , and $\bar{\beta}_{p,rms}$ is the RMS BW of $s_{p,n}$. The β_r used to calculate α_{wb} through (9) and (10) can be found from the FE manual.

According to (27) and (28), the carrier-phase tracking error for the pilot channel can be estimated by:

$$\delta \hat{\theta}_p = \tan^{-1} \left[\frac{Q_p(0)}{I_p(0)} \right] \quad (32)$$

Then, the optimal weighed averaging of $\delta \hat{\theta}_d$ and $\delta \hat{\theta}_p$ can be obtained as:

$$\delta \hat{\theta}_{wb} = \frac{1}{4} \cdot \delta \hat{\theta}_d + \frac{3}{4} \cdot \delta \hat{\theta}_p \quad (33)$$

This is because $\xi_d : (\xi_{p11} + \xi_{p61}) = 1 : 3$.

Using the captured signal, four ideal ACF magnitude curves shown in Fig. 12 are produced by shifting their corresponding spreading waveform replicas from the prompt code phase obtained from the WB tracking mode. As can be seen, except the ACF for $s_{p61,n}$ that typically cannot be used alone for positioning, the complete pilot channel has the sharpest ACF main peak, implying excellent performance of the QMBOC spreading waveform in terms of ranging accuracy and multipath resistance. This is in line with the largest $\bar{\beta}_{rms}$ of QMBOC for a fixed β_r , as shown in Fig. 2.

Tracking quality parameter calculation

In the SDR, C/N_0 estimation and PLL lock detector output are generated to indicate signal tracking quality. The C/N_0 estimation is based on the variance summing method (Sharawi et al. 2007). Figure 13 shows the C/N_0 estimations for data and pilot channels in WB tracking mode. The

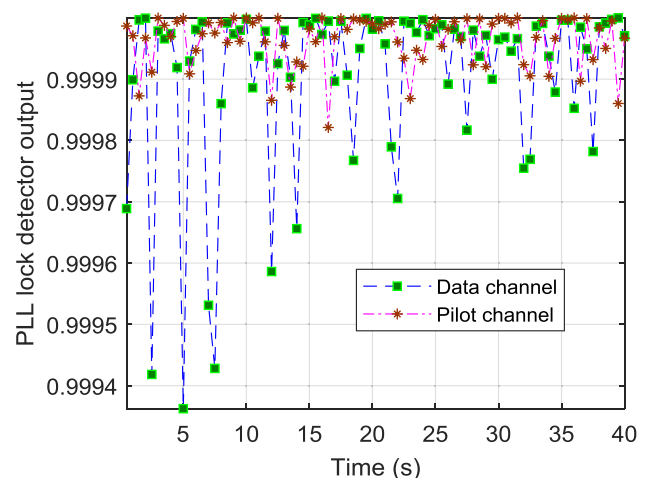


Fig. 14 PLL lock detector outputs for PRN #20 in WB tracking mode

Fig. 15 Encoding and decoding orders of the B-CNAV1 message (Li et al. 2018)

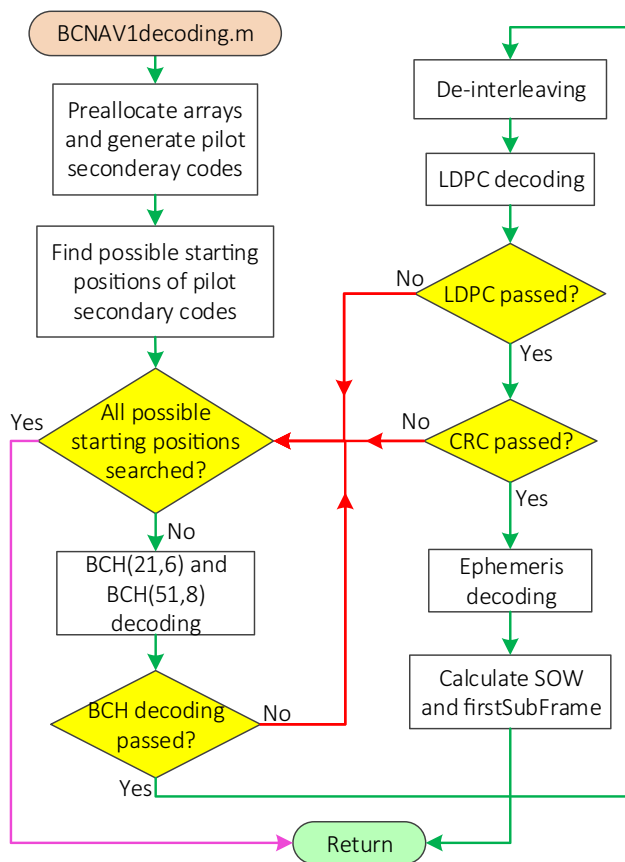
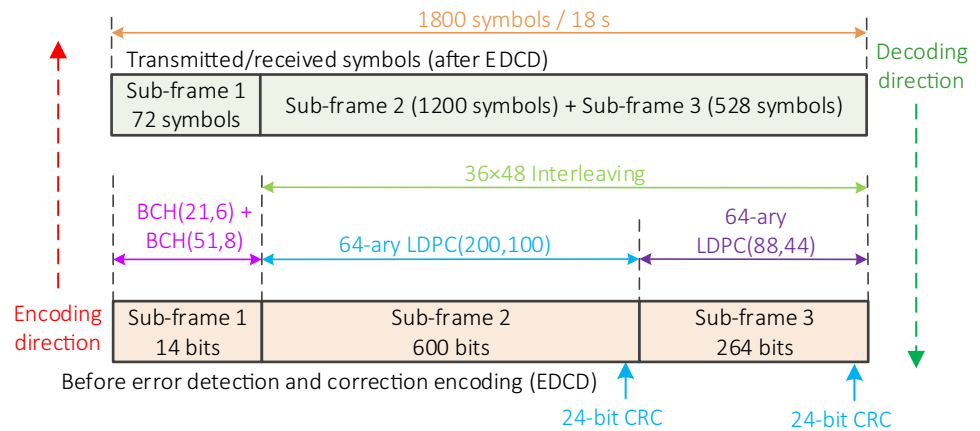


Fig. 16 Flow diagram of the B-CNAV1 message demodulation (Li et al. 2018)

averaged C/N_0 of the pilot channel is 4.746 dB-Hz higher than that of the data channel, which is close to the theoretical value of $10 \cdot \log \left[(\xi_{p11} + \xi_{p61}) / \xi_d \right] = 4.77$ dB-Hz.

Denoted as L_d , the PLL lock detector output is a function of phase tracking error, $\delta\theta$, satisfying $L_d = \cos(2\delta\theta)$ (Van Dierendonck 1996). This means in good lock conditions, the L_d is nearly 1. Figure 14 shows the L_d in WB tracking mode. As can be seen, the L_d values for both channels are greater

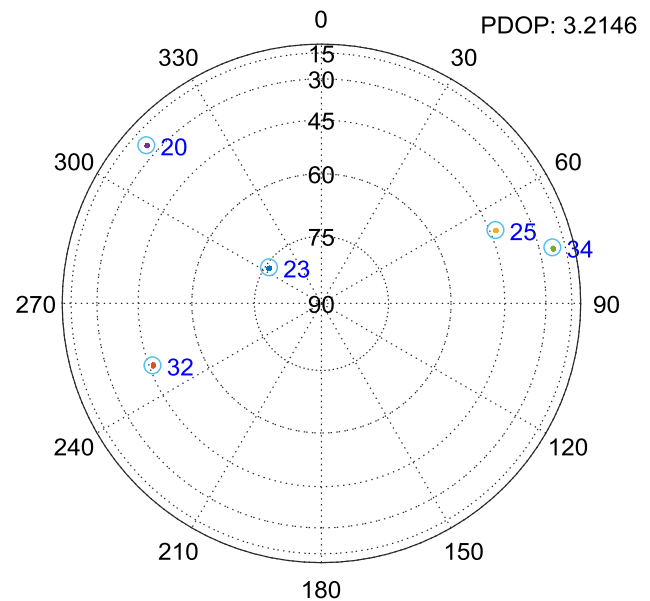


Fig. 17 Sky-plot of the acquired BDS-3 satellites

than 0.999, indicating accurate carrier-phase tracking. On average, the L_d for pilot channel is greater than that for the data channel. This is because the prompt correlation values of the pilot channel used to calculate L_d is less affected by noise due to the larger power allocated to pilot channel.

NAV message demodulation

The NAV message of the B1C signal adopts the B-CNAV1 encoding format defined by CSNO (2017c) which includes a set of advanced error detection and correction encoding (EDCD) techniques as shown in Fig. 15. The 14 bits of the PRN and the seconds of hour (SOH) message in subframe 1 are encoded into 72 symbols using a combined Bose–Chaudhuri–Hocquenghem (BCH) encoding. After encoded through the 24-bit cyclical redundancy check (CRC-24Q), subframe

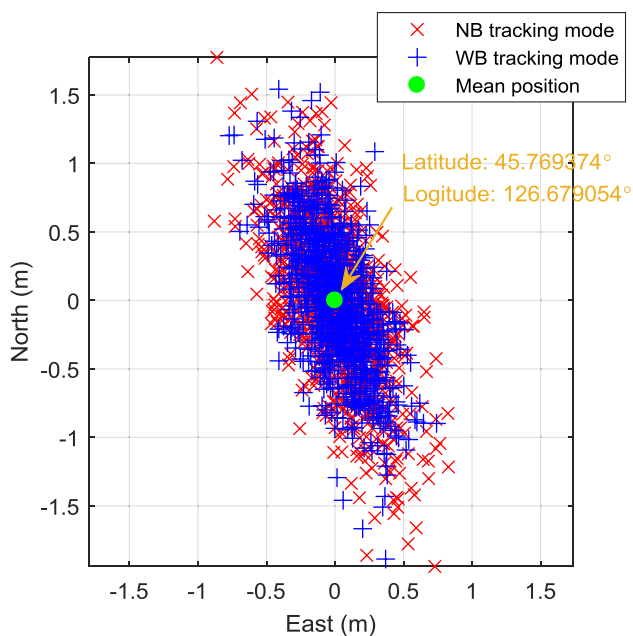


Fig. 18 Dispersion of the horizontal position coordinates computed from NB and WB tracking modes

Table 1 STDs of the 3D position coordinates computed from the two tracking modes

Tracking modes and STD percentage decrease	STD σ (m)		
	East	North	Up
NB tracking σ_{NB}	0.275	0.587	0.774
WB tracking σ_{WB}	0.225	0.503	0.611
$(\sigma_{NB} - \sigma_{WB})/\sigma_{NB}$ (%)	18.3	14.3	21.0

2 and subframe 3 are separately encoded using rate 1/2 low-density parity-check (LDPC) codes; symbols of these two subframes are then combined and interleaved using a block interleaver to combat bursts of errors. After EDCD, the original 878 message bits in a B-CNAV1 frame are converted to 1800 symbols for broadcast. These powerful EDCD techniques allow the ephemeris to be read at low C/N_0 levels (Betz et al. 2006a, b), which is a main enabler for allocating 75% power to the B1C pilot channel. The extracted message symbols from the PLL tracking must be decoded in an inverse process of the encoding.

Since the boundaries of the message symbols align with the primary codes of the data channel, no symbol/bit sync is required for the B-CNAV1 message demodulation. Therefore, the first operation for B-CNAV1 message demodulation is the frame sync. Because the B-CNAV1 message does not provide any preamble sequences, frame sync cannot be performed as for the legacy GNSS signals. However, as the

pilot-channel secondary code's length is the same as that of the B-CNAV1 frame, synchronization to the secondary code achieves alignment to the B-CNAV1 frame. Based on this approach, the flow diagram for the B-CNAV1 data demodulation is illustrated in Fig. 16.

PVT computation

After the announcement of the basic BDS-3 constellation deployment on December 27, 2018, live B1C signals were captured from a static position to test the PVT solution calculation of the developed SDR. Figure 17 shows the sky-plot of five acquired BDS-3 satellites. Represented in the local geodetic horizon system, the dispersion of the horizontal position coordinates computed from NB and WB tracking modes are presented in Fig. 18. The standard deviations (STDs) of the 3D coordinates are listed in Table 1. It shows that the WB tracking mode decreases the position SDs along the east, north and up directions by 18.3%, 14.3%, and 21.0%, respectively, compared to the NB tracking mode. These percentage decrease values are close to the theoretical value of 17% calculated through (11) when the FE BW $\beta_r = 33$ MHz.

Conclusion

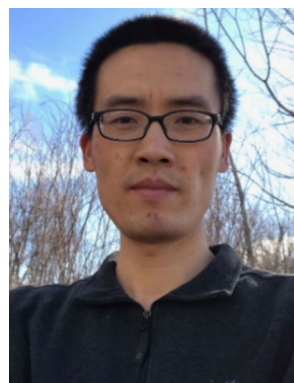
The BDS-3 B1C/B2a SDR development is described, and the primary processing results of captured signal are reported, with emphasis on the B1C processing. The SDR is able to perform complete functionalities from acquisition through PVT calculation. Separate or joint processing of data and pilot channels for acquisition and tracking are implemented for B1C and B2a signals. This provides basic methodologies for the B1C QMBOC signal processing based on NB or WB correlation approach. The implementation detailed in this work is not only expected to help the GNSS community to understand the latest BDS-3 open-service signals quickly but also to enable advanced algorithm development on the existing SDR. For example, due to the similarity of the QMBOC to TMBOC, the SDR can easily be modified to processing/analyzing the Quasi-Zenith Satellite System (QZSS) L1C signals and the GPS L1C signals.

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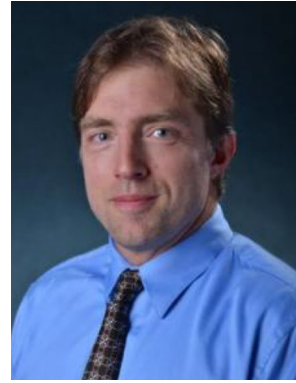
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